

PTVA's

MULUND COLLEGE OF COMMERCE (AUTONOMOUS)

Programme: B.Sc. Information Technology

Class: TY

Semester: V

Exam: ESE

Name of the Course: ASP.NET Core

Date of Exam: 09/10/2025

Marks: 60

Time: 2 hours

Instructions:

- (1) All questions are compulsory.
- (2) Make suitable assumptions wherever necessary and state the assumptions made.
- (3) Answers to the same question must be written together.
- (4) Numbers to the right indicate marks.
- (5) Draw neat labeled diagrams wherever necessary.
- (6) Use of non-programmable calculator is allowed.

**Q 1 Attempt any TWO of the following:**

12

- A. Describe the role of the ASP.NET Core Platform in application development by outlining how it supports building, running, and maintaining modern applications.
- B. List and explain the steps involved in creating an ASP.NET Core project using Visual Studio or the CLI.
- C. Illustrate with example how N unit testing is implemented in ASP.NET Core applications.
- D. Examine the case where a developer forgets to configure launchSettings.json
  - i) What issue will occur at runtime?
  - ii) Suggest possible remedies.

**Q 2 Attempt any TWO of the following:**

12

- A. State pagination in Blazor and mention its use in eCommerce applications.
- B. Interpret the situation where the shopping cart resets after page refresh.
  - i) Identify the reason.
  - ii) Suggest a solution to persist cart data.
- C. Summarize the process of adding navigation controls in Blazor.
- D. Demonstrate with example how a service class manages cart functionality in Blazor.

**Q 3 Attempt any TWO of the following:**

12

- A. Describe how custom middleware can be created and configured in ASP.NET Core.
- B. Analyze the output generated by the following middleware code:

```
app.Use(async (context, next) => {  
    await context.Response.WriteAsync("Start1 ");  
    await next();  
    await context.Response.WriteAsync("End1 ");  
});  
app.Use(async (context, next) => {  
    await context.Response.WriteAsync("Mid2 ");  
});
```

  - i) Break down the sequence of output produced.
  - ii) Examine the significance of next() in determining the flow of execution.

- C. State the purpose of the Identity Database in ASP.NET Core with an example.
- D. Demonstrate with code snippet how Dependency Injection is used for service management.

**Q 4 Attempt any TWO of the following:**

12

- A. Define response caching in ASP.NET Core and state its syntax for implementation.
- B. Explain the role of Entity Framework Core in data handling within ASP.NET Core.
- C. Evaluate the effectiveness of RESTful Web Services compared to traditional SOAP services.
- D. Analyze the case where a developer receives a 500 Internal Server Error for POST requests in an API.
  - i) Determine the possible reason for this issue.
  - ii) Examine how error-handling middleware can address the problem.

**Q 5 Attempt any TWO of the following:**

12

- A. Identify what Tag Helpers are in ASP.NET Core and describe how they are used.
- B. Describe the concept of Razor Pages routing in ASP.NET Core with an example.
- C. Examine the situation where a developer uses ViewBag to transfer data, but the value disappears after redirect.
  - i) Why does this happen?
  - ii) Suggest an alternative method to retain data.
- D. Develop a simple View Component in ASP.NET Core to render a list of featured products.